

Please group together with your Operating System

MacOS?

-> Please sit to the left

Linux?

-> Please sit in the middle

Windows?

-> Please sit to the right

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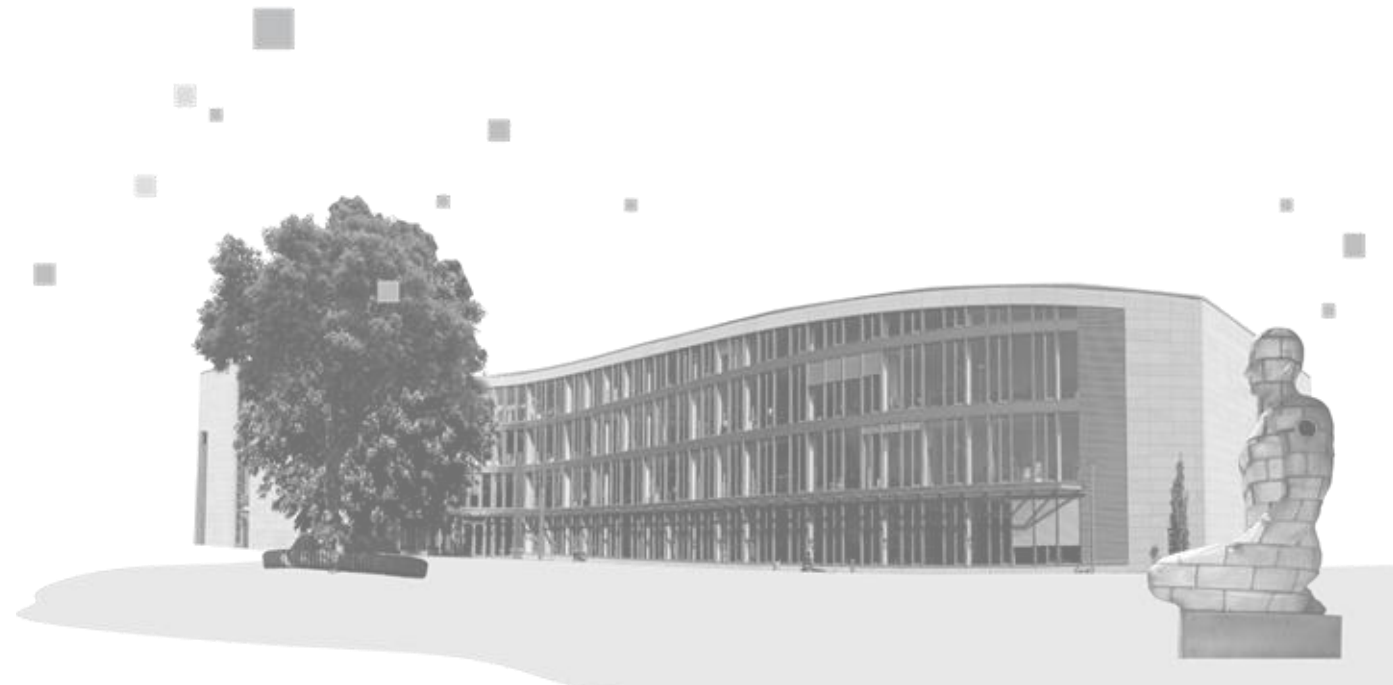


First Steps Towards Your Own AI Application

Setting up the key tools for local AI development

**Design IT.
Create Knowledge.**

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Overview of the AI Service Center

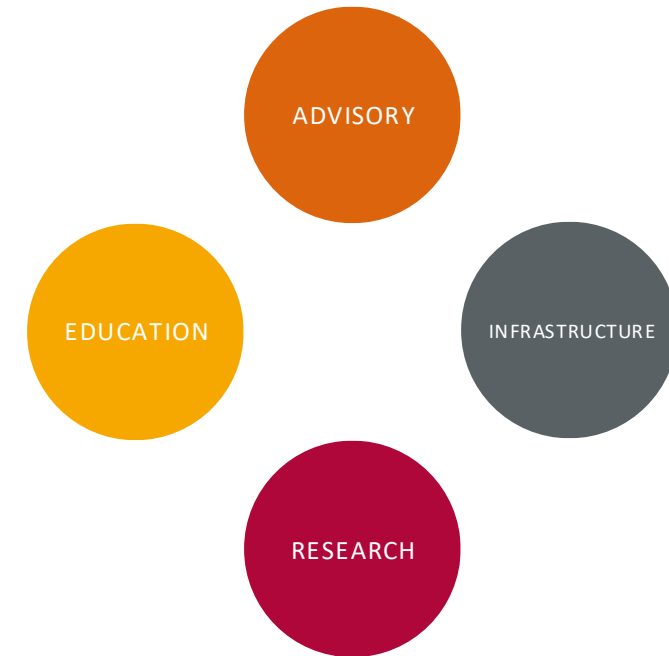
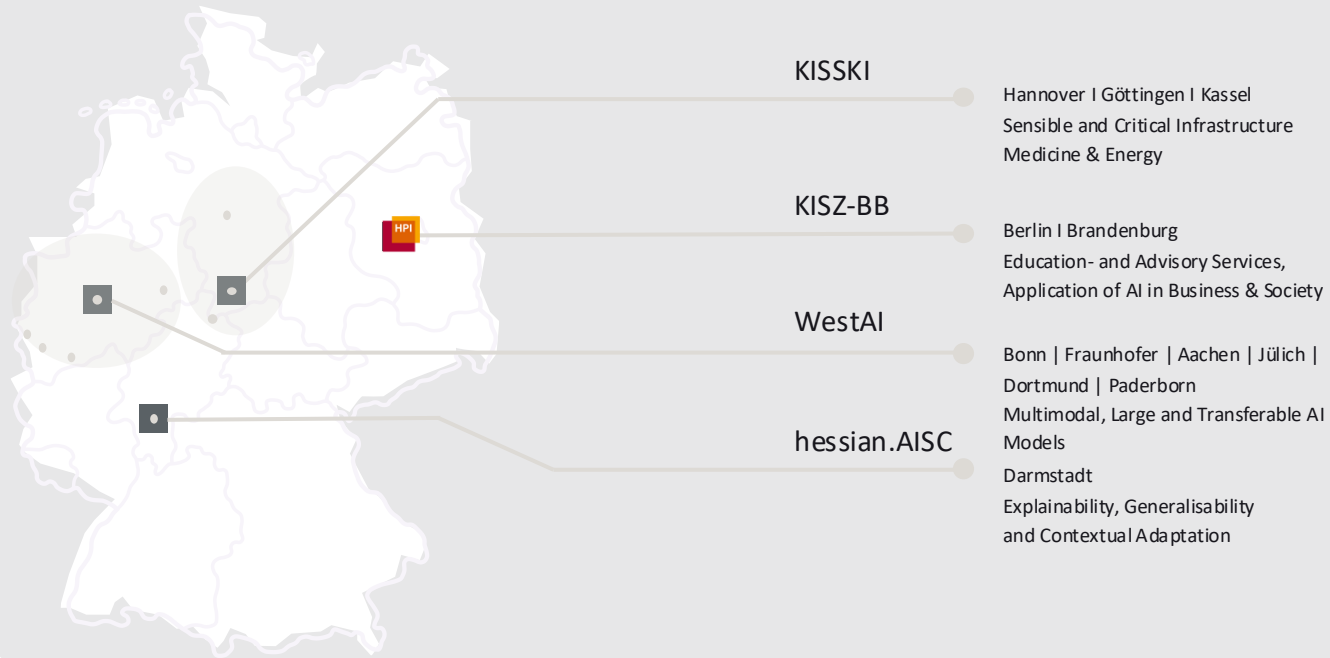


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AI-Service Centers



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ADVISORY

AI Advisory

- General Questions regarding AI,
- Computing Infrastructure,
- Model, Architecture, Framework Selection

AI Pilot Projects (3 months)

1. Understanding and Identifying the Use Case
2. Specifying the Use Case
3. Developing a Solution
4. Finalizing and Documenting

Cooperations

- Co-organized Network Meetings or Workshops

Start-up Support

- All of the above
- Support with Networking

HPI ECOSYSTEM



Infrastructure



Training

- 8 x 8 NVIDIA H100 GPU

Inference

- 5 x 8 NVIDIA A30 GPU

ARM Server

- Ampere Altra Max M128-30 CPU
- 2 x NVIDIA L40 GPUs

GPU Server

- AMD Epyc CPU
- 8 x NVIDIA L40S GPU

Edge

- ARMv8 CPU
- NVIDIA Jetson AGX Module

Storage

- 1.5 PB NVRAM

Network

- 400 Gb/s Infiniband
- 200 Gb/s Ethernet

kisz@hpi.de

hpi.de/kisz

EDUCATION & TRAINING

Talks



Guest Talk: Automatische Deepfake-Erkennung mit KI

Di. 17. März, 17:00

Hasso-Plattner-Institut

Kostenlos

MOOCs



Chatbots leicht gemacht: Große Sprachmodelle einfach nutzen

📅 SELF-PACED SINCE DEC 10, 2025

DATA SCIENCE

🏆 RECORD OF ACHIEVEMENT

🌐 DE

Workshops

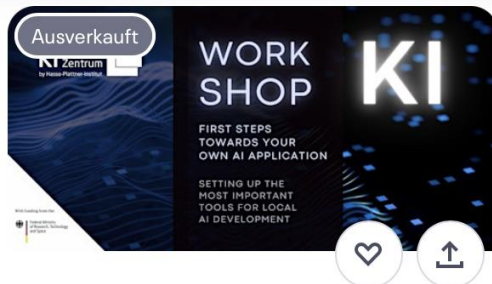


Reinforcement Learning – Part II

Do. 26. Feb., 10:00

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First Steps Towards Your Own AI Application

Donnerstag um 15:00

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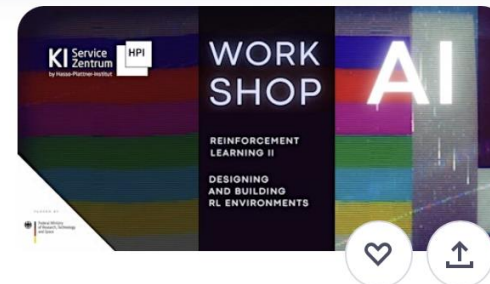


Effizientes Abfragen von Informationen – Teil II

Mi. 25. Feb., 13:00

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Reinforcement Learning – Part II

Do. 26. Feb., 10:00

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Erste Schritte zu Ihrer eigenen KI-Anwendung

Fr. 13. März, 13:00

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Guest Talk: Automatische Deepfake-Erkennung mit KI

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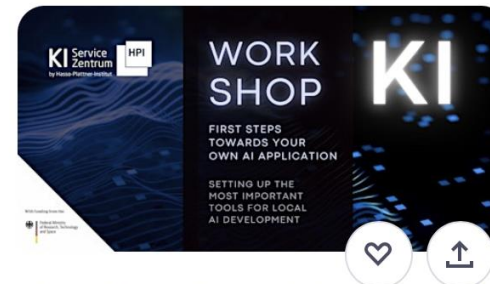


Effizientes Abfragen von Informationen – Teil III

Do. 26. März, 14:00

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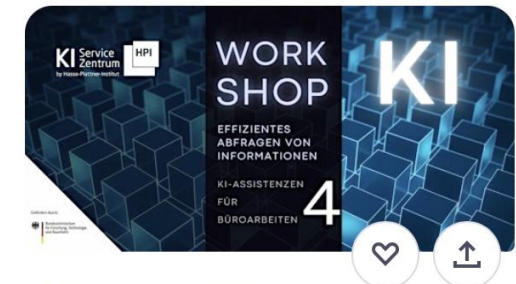


First Steps Towards Your Own AI Application

Mo. 13. Apr., 10:00

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Effizientes Abfragen von Informationen – Teil IV

Di. 28. Apr., 15:00

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Goal: Set up your environment

1. Installation

- a) VS Code
- b) GitHub
- c) Python
- d) UV
- e) Jupyter
- f) Docker Desktop
- g) Ollama

2. Outlook

- Workshops by the AI Service Center

Let's get to know each other

- What is your background?
- What is your motivation coming here today?
- Are you familiar with VSCode, Git, Docker, Python, ... ?

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Overview of the software

AISC Getting Started - AI Chatbot Demo

Chat with a local AI model powered by Ollama

AI Chat Interface

● Connected to Backend   Clear



Hello! I'm an AI assistant powered by Ollama running locally. Ask me anything!

which model are you?



I'm Llama 3.2 1B. It's a large language model developed by Meta, designed to provide helpful and informative responses to your questions.

Are you running locally on my laptop?



Yes, I'm running locally on your device. This means that the AI was trained on a dataset of text from the internet, which is then used to generate human-like responses to your queries. Your language input is then passed through my network of algorithms and models, which produce the output you see.

Type your message here... (Press Enter to send)



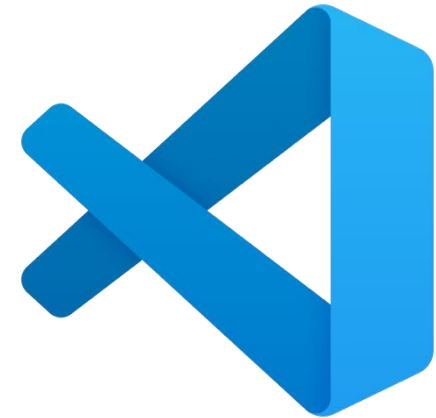
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Visual Studio Code (VS Code)

- Code-Editor:
 - Lightweight
 - Cross-platform
 - Expandable with extensions
- Supports various tools:
 - Python, Jupyter Notebook, Git, Docker, SSH
- Integration of coding assistants (e.g., Claude)



GitHub & Git

- Platform for version control of code
 - Development, version control, review, and documentation



With funding from the:



Python

- General-purpose programming language with key libraries for AI and data science
 - NumPy: numerical computing
 - PyTorch: neural networks
 - Librosa: audio processing
 - ...
- Extensive learning resources and tutorials available
- Suitable for data processing, model training, ...



UV

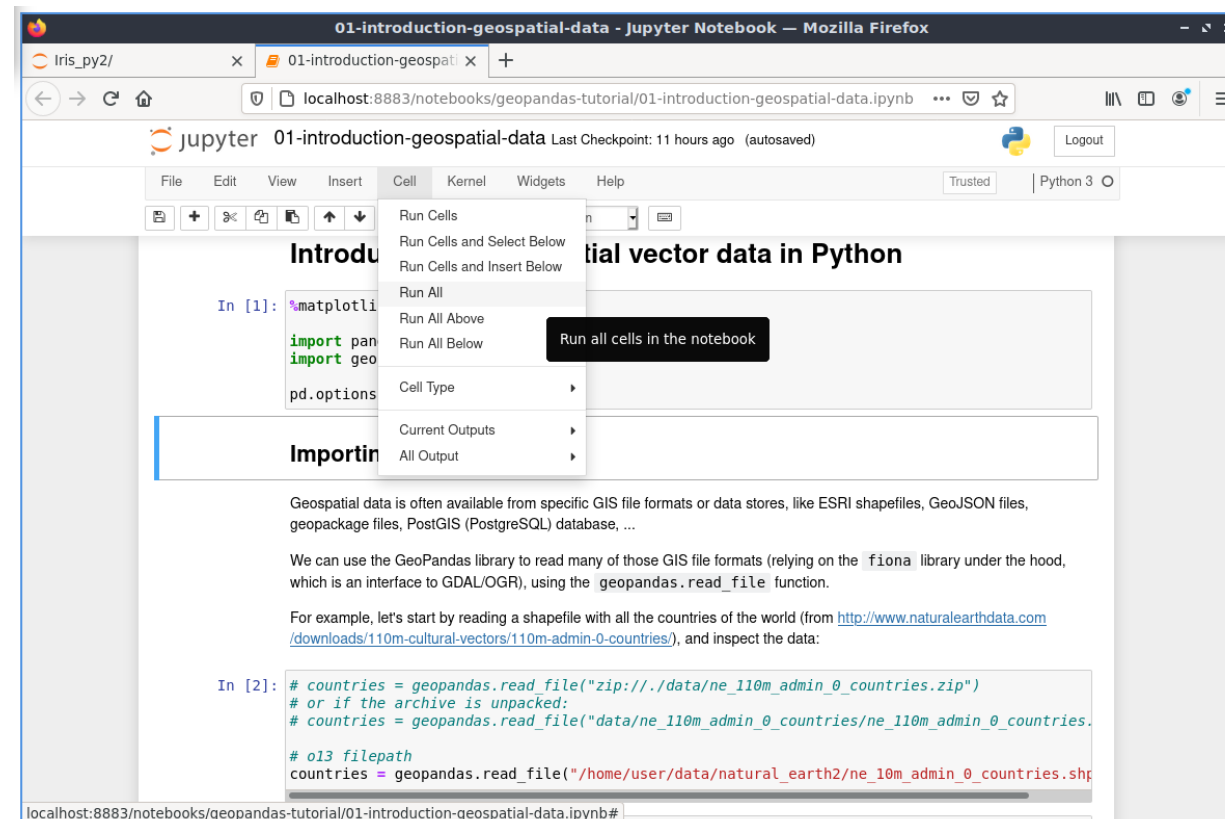
- Fast Python package and project manager
 - Handles package installation and dependency management
 - Creates reproducible development environments, ensuring consistent package versions
- Ensures code runs reliably across different machines and operating systems
 - Windows 10, 11, XP
 - Linux Mint, Ubuntu, Debian
 - MacOS



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Jupyter notebooks

- Supports code, narrative text, equations (Markdown), and visualizations
- Ideal for exploratory data analysis, prototyping, and teaching
- Code cells can be executed independently --> Easy debugging



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Docker

- Container platform for packaging reproducible runtime environments
 - Enables applications to run consistently across machines, servers, and high-performance clusters (HPC)
 - Solves "works on my machine" problem
- Ideal for complex software stacks



With funding from the:

<https://github.com/aihpi/workshop-getting-started>

 README



Gefördert durch:



Bundesministerium
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und Raumfahrt

AI for Everybody - Getting Started

This repository provides a comprehensive getting started guide for people new to development workflows. Learn how to set up your development environment, work with Python virtual environments, Docker containers, and AI tools like Ollama - all from scratch!

Workshop Context

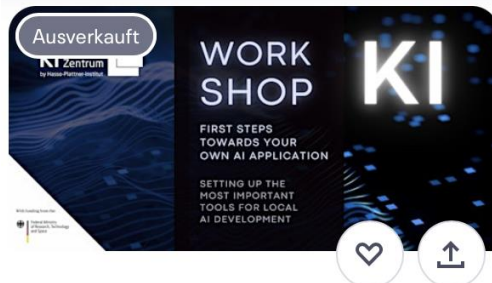
This repository introduces the essential development tools and workflows used in workshops offered by the **AI Service Centre Berlin Brandenburg**. Workshop participants should complete the installation process before attending workshops. The installation can be completed within one hour. Because the installation process involves downloading multiple GB of software, slow internet connections can lead to longer installation times.

Need assistance? If you encounter any issues during installation, please contact us at kisz@hpi.de for technical support.

What You'll Learn

- **Development Environment Setup:** Install and configure VS Code, Git, and Python
- **Python Environment Management:** Master UV for creating and managing virtual environments
- **Jupyter Notebooks:** Learn to use Jupyter for interactive development
- **Docker Containers:** Build and run applications in containerised environments
- **AI/LLM Integration:** Set up and use Ollama for local language model development
- **Simple Application:** Build a simple chatbot that uses local AI models

Finished? Check out our upcoming workshops, talks and online courses!



First Steps Towards Your Own AI Application

Donnerstag um 15:00

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Effizientes Abfragen von Informationen - Teil II

Mi. 25. Feb., 13:00

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Reinforcement Learning - Part II

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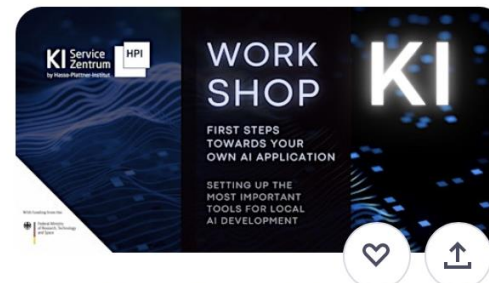


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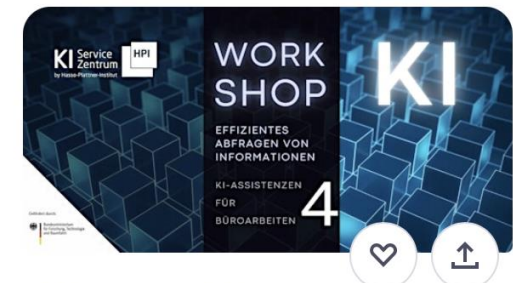


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Podcast

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Digitale Gerechtigkeit - Wie du dich für eine fairere KI-Zukunft engagieren...

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Efficient AI for Weather Forecasting

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BIG DATA AND AI

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Chatbots leicht gemacht: Große Sprachmodelle einfach nutzen

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Lehrkräfteprofessionalisierung für KI in Schule und Unterricht

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PROFESSIONAL SKILLS

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Argumentieren lernen mit KI - für Schule & Alltag

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Time Series Analysis

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Sustainability in the Digital Age: Efficient AI Techniques in the LLM Era

SELF-PACED SINCE JUN 10, 2025

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Practical Computer Vision with PyTorch

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KI-Biases verstehen und vermeiden

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Profitable AI

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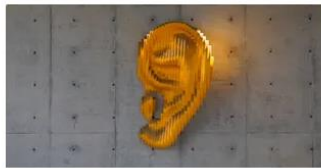
Understanding Embeddings for Natural Language Processing

SELF-PACED SINCE DEC 24, 2023

BIG DATA AND AI, DATA SCIENCE

CONFIRMATION OF PARTICIPATION

EN



Selection and Implementation of AI Models: Speech2Summary

SELF-PACED SINCE SEP 30, 2023

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form



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